Intelligent Medical Report Analysis System

Project Domain: DL / GenAI / NLP

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Problem Definition & Target Objectives

Clinical Problem & Jargon Barrier

Medical lab reports use complex medical terms and shorthand, making them difficult to understand and increasing patient anxiety, confusion, and risk of misinterpretation.

Generic AI Hallucination & Cloud PHI Risk

LLMs can produce unstructured text, hallucinate critical values, and raise HIPAA/GDPR privacy concerns when PHI is sent to public cloud APIs.

Objective	Target	Status Achieved
Extraction Accuracy	Entity F1 ≥ 0.85	0.9991 (EXCEEDED)
Abnormality Flagging	Deterministic ranges	ZERO HALLUCINATION
Readability Level	FKGL below 8	GRADE 8.4 (MEAN)
Output System	Dual Output	DASHBOARD + NARRATIVE

Scope Boundaries & Market Benchmarks

In-Scope Functionality

Process lab reports (PDF/Image), extract entities (test, value, unit), flag against reference bounds, plain-English summary, local offline execution.

Out-of-Scope Boundaries

No clinical diagnosis/prescriptions, no medical imaging (X-Ray/CT/MRI), no handwritten notes. Proof-of-concept for educational comprehension.

Platform	Key Strengths	Core Limitations
Med-PaLM	High medical QA accuracy	Proprietary & Closed-Source
MedGemma	Multimodal health understanding & open weights	Requires clinical fine-tuning for lab extraction
Glass Health	Differential diagnosis	Clinician focus, no patient UI
UpToDate / AMBOSS	Trusted medical databases	Manual lookup, requires clinical background
Our System	Automated, 100% private	GPU dependency

Problem Solution – ClinicalBERT Extraction and BioMistral-7B for Zero Hallucinations and generating Patient-Friendly Insights

STEP 01
Image / PDF Input

STEP 02
OCR + Preprocessing

STEP 03
ClinicalBERT NER Extraction

STEP 04
Deterministic Abnormality Check

STEP 05
BioMistral-7B (QLoRA) Explain

STEP 06
Dashboard + Patient Summary

- **ClinicalBERT** extracts lab entities and converts them into structured JSON, keeping extraction separate from text generation.
- A rule-based module compares values with reference ranges to classify results as HIGH , LOW or NORMAL .
- **BioMistral-7B** only converts the verified structured results into patient-friendly explanations, **reducing numerical hallucination**.

Dataset Preparation & Corpus Curation

Patient-Level Split (Zero Leakage)

Exploratory Data Phase
Started with Kaggle MIMIC-III sample (100 patient records, 76,074 LABEVENTS rows) + MTSamples (2,416 lab transcriptions).

Full PhysioNet Access & Scaling
Expanded to full PhysioNet access (1.65M records). Ablation studies proved a 20,000 stratified report subset delivered identical F1 accuracy while reducing training time from >10 hrs to 30 mins.

MTSAMPLES EXTRACTION

1,630 Unique

Total MTSamples Reports **4,999 total**
Lab-Related Keywords **582 keywords**
Lab Reports Identified **2416 lab tests related**
Deduplicated Reports **1,630 unique**

FULL MIMIC-III MINING

Total 1.6 Million Reports

Total LABEVENTS Rows **27.85M events**
Valid Numerical Results **24.93M valid**
Patients with Lab Data **46,250 patients**
D_LABITEMS Rows **753 (589 unique)**
Structured Lab Entities **231,988 entities**

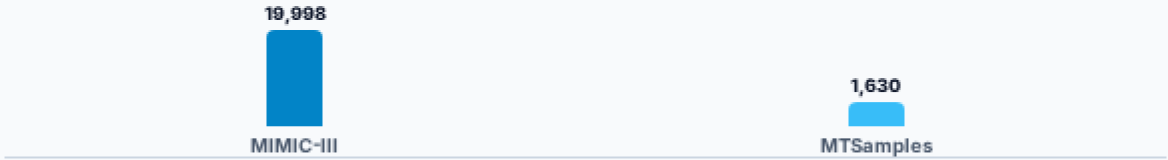
FINAL CORPUS (70/15/15)

21,628 Reports

MIMIC Synthetic Share **19,998 (92.5%)**
MTSamples Share **1,630 (7.5%)**
Training Set (70%) **15,138 reports**
Validation Set (15%) **3,245 reports**
Test Set (15%) **3,245 reports**
Patient Leakage Audit **Passed (0)**

TRAIN / VALIDATION / TEST SPLIT (70/15/15)

DATASET SOURCE DISTRIBUTION



END TO END DATA PROCESSING PIPELINE

1. Raw Datasets 4.9k MTSamples / 27.8M MIMIC › **2. Lab Filtering** 582 Keywords / 589 LABITEMS › **3. Deduplication** 1,630 Unique MTSamples › **4. Synthetic Generation** 19,998 MIMIC Reports › **5. Total Corpus** 21,628 Reports › **6. Patient Split** 15.1k / 3.2k / 3.2k

NER Annotation & Tokenization for ClinicalBERT

Dictionary preparation, BIO tag classification scheme, and token distribution plots

2.4M Tokens Annotated

1. Dictionary & Annotation

589	1,859
MIMIC-III LABELS	SEARCHABLE VARIATIONS

Rule-based annotation engine

Mimic Labels + regex patterns auto-labelled medical entities across the corpus - replacing slow manual labelling.

Tag-Set Refinement

9 IOB tags → pruned to 5: {B-TEST, I-TEST, B-VALUE, B-UNIT, O}.

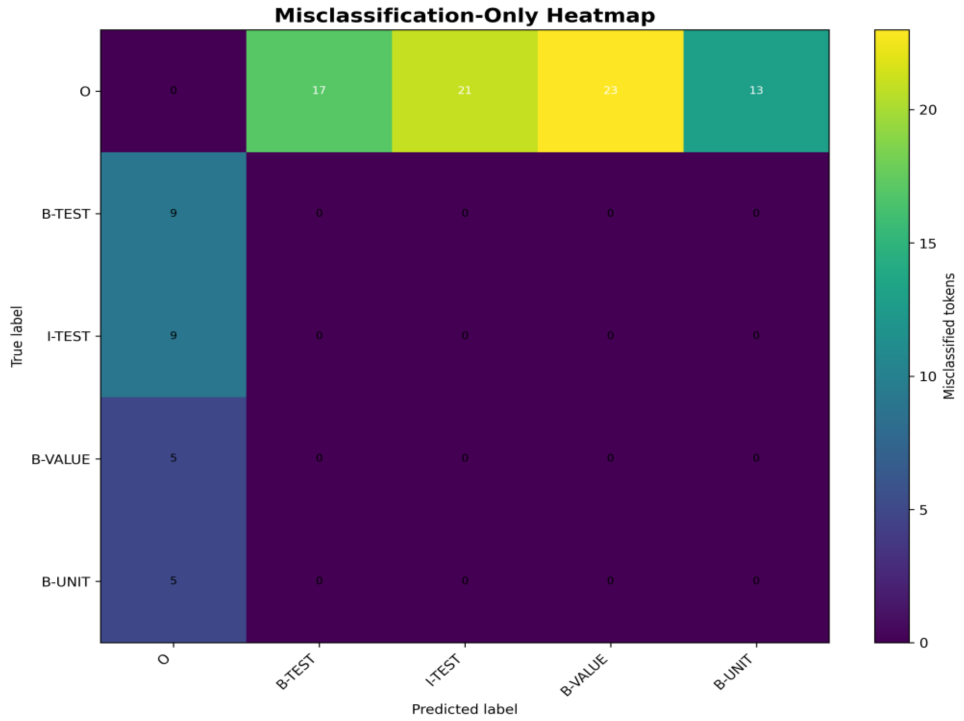
Fixed class imbalance & unit/range confusion via loss re-weighting.

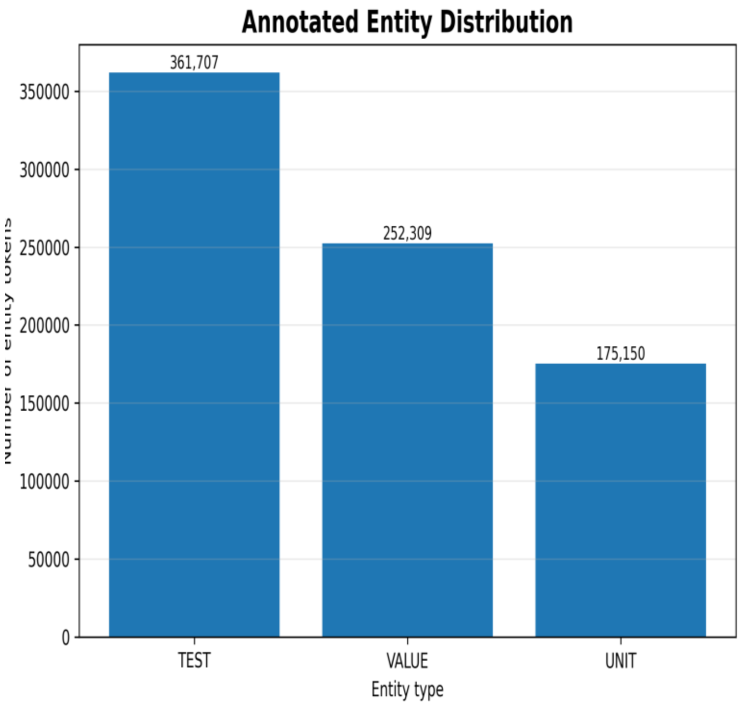
Switched model selection to Entity F1 (not token accuracy).

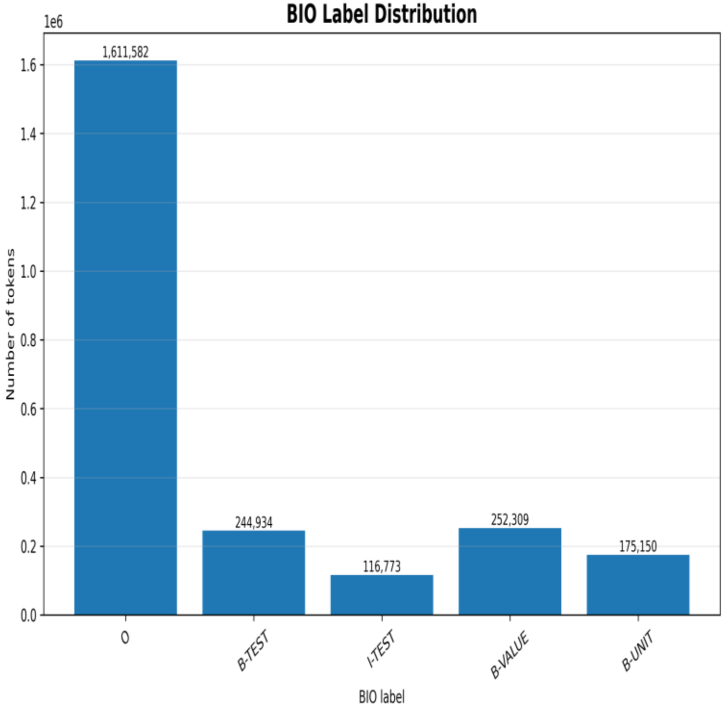
2. NER Labels & Concrete Example

Label	Meaning / Entity Category
B-TEST	Beginning of test name
I-TEST	Continuation of test name
B-VALUE	Numerical laboratory result
B-UNIT	Laboratory measurement unit
O	Other / non-entity token

Anion	Gap	=	22	mEq/L
B-TEST	I-TEST	O	B-VALUE	B-UNIT







System Architecture Design & Pipeline

1. Input
PDF / Image Scan

2. Stage 1 OCR
PaddleOCR + Preprocess

3. Stage 2 NER
ClinicalBERT (JSON)

4. Stage 3 Check
Deterministic Engine

5. Stage 4 LLM
BioMistral-7B QLoRA

ClinicalBERT — NER Extractor Model Specs

Architecture: 110M params · 12 layers · 768 hidden · 12 heads

Pretraining Domain: Pretrained on MIMIC-III clinical notes

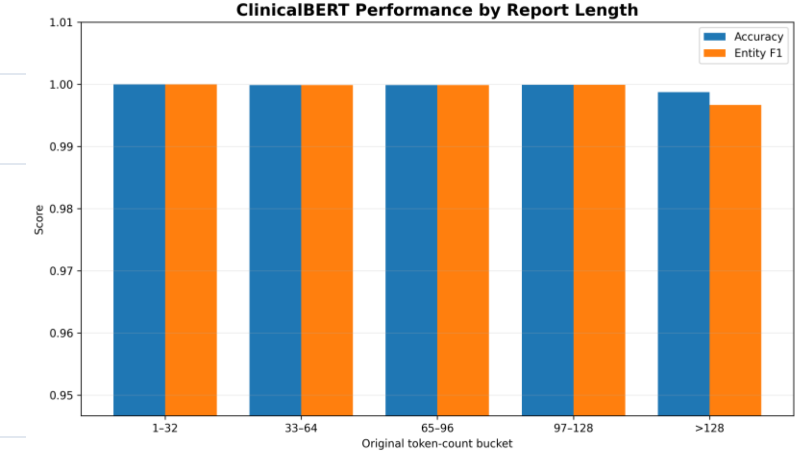
Task Head: Token classification head (5 BIO tags)

Hyperparameters: lr 2e-5 · 10 epochs · Best @ Epoch 9 (val loss 0.0016)

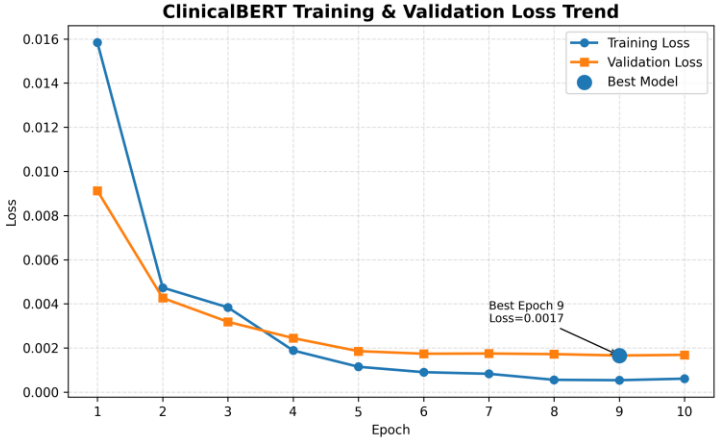
Extraction F1 Score: 99.91% (entity-level)

ClinicalBERT Training Metrics 10 Epochs Evaluation

Epoch	Train Loss	Val Loss	Accuracy	Entity F1
Epoch 1	0.01584	0.00912	0.9979	0.9979
Epoch 3	0.00384	0.00318	0.9993	0.9993
Epoch 5	0.00115	0.00185	0.9996	0.9996
Epoch 9 BEST	**0.00054**	**0.00166**	**0.9997**	**0.9991**



Dataset Size	Train Time	Entity F1	Decision
1.6M	>10 h	99%	Slow
21628	**0.5 h**	**99.91%**	**Selected**
5,000	0.1 h	98.2%	Slight drop



Real-World Images/PDF OCR Challenges & Fixes

VISION OCR PIPELINE

Challenge - Zero Lab Tests Identified using Fine-Tuned ClinicalBert on Raw Text extracted from Lab Tests Images/PDF using OCR

Fixes

1. Rotation Detection & Correction

Canny + Hough Lines detect and correct image rotation.

2. Image Preprocessing

Upscaling, CLAHE, and sharpening enhance image quality.

3. PaddleOCR Extraction

Extracts text and bounding boxes from the report.

4. Bounding-Box Line Sorting

Reconstructs OCR tokens into correctly ordered text lines.

5. Value-Line Filtering

Keeps relevant laboratory lines containing values and test keywords.

6. Lexicon Normalization

Corrects OCR misspellings using medical test-name patterns.

Real-World Image Format Evaluation

Average Detection & Accuracy Rate (%)

DARSS Composite Metric & Success

COMPOSITE EVALUATION FORMULA

$$DARSS = (0.45 \times Text_{Det}) + (0.35 \times Val_{Acc}) + (0.20 \times Unit_{Acc}) + Bonus$$

85%	0%	15%
FULL SUCCESS	**PARTIAL**	**UNSUCCESSFUL**
DARSS ≥ 70%	DARSS 50–69%	DARSS < 50%

Overall Error Rate

Text Detection Error 15.3%

15.3%

Value Extraction Error 14.8%

14.8%

Unit Extraction Error 22.2%

22.2%

Overall Model Performance

Precision	98.2%
Recall	83.8%
F1-Score	89.3%
Value Accuracy	85.2%
Unit Accuracy	77.8%

Format Evaluation Breakdown

Image Format Type	Count	Det %	Val %	Unit %
Screenshot	3	100.0%	100.0%	86.1%
Rotated Image	4	88.2%	88.2%	72.7%
PDF Report	2	81.8%	90.0%	70.0%
Human Captured	3	77.0%	77.0%	87.7%
Low-Quality Image	3	76.9%	87.0%	69.8%
Standard Lab	5	69.0%	69.0%	61.7%

System Architecture Design & Pipeline

1. Input	2. Stage 1 OCR	3. Stage 2 NER	4. Stage 3 Check	5. Stage 4 LLM
	PaddleOCR + Preprocess	ClinicalBERT (JSON)	Deterministic Engine	BioMistral-7B QLoRA

BioMistral-7B QLoRA Explainer Model Specs & Metrics

- **Quantization:** 4-bit (NF4 + double quant) · bfloat16
- **LoRA Config:** Rank r=8, α=16 · 20.97M params (0.2888%)
- **Selection:** Outperformed Mistral, Phi-3, Gemma-3, Qwen-3

Metric	Value	Interpretation / Note
Eval NLL Loss	0.6906	**Lowest loss among 8 trials**
Perplexity	1.99	eNLL · High token certainty
Eval Token Acc.	80.75%	Agreement with reference text
Train-Eval Gap	0.0587	0.6906 - 0.6319 · Strong general.

Lowest Zero-Shot FKGL + 100% Numeric Fidelity

8-Trial QLoRA Exploration Matrix Selected Candidate: Trial 5

Trial #	Rank (r)	Alpha (α)	LR	Train Loss	Eval Loss	Token Acc.	Status / Finding
Trial 0	16	16	1.05e-4	0.8808	0.7697	78.54%	Moderate perf.
Trial 1	32	32	7.79e-5	0.8047	0.7427	79.35%	Slower conv.
Trial 2	32	128	2.18e-3	0.8324	0.7847	78.52%	Underfit (low LR)
Trial 3	64	256	2.36e-4	0.5425	0.7403	80.33%	Overfit (high cap)
Trial 4	32	64	2.75e-3	0.9491	0.8515	76.80%	Highest eval loss
Trial 5 (Best)	8	16	3.31e-4	0.6319	0.6906	80.75%	Optimal balance
Trial 6	16	64	2.56e-4	0.5168	0.7131	81.07%	Slight overfit
Trial 7	32	128	2.18e-5	0.8331	0.7851	78.55%	Underfit (low LR)

100%
Numeric Fidelity
Every value, unit & status reproduced exactly

0.52
ROUGE-L
Strong content overlap with reference summaries

0.40
BLEU
Respectable n-gram precision for open generation

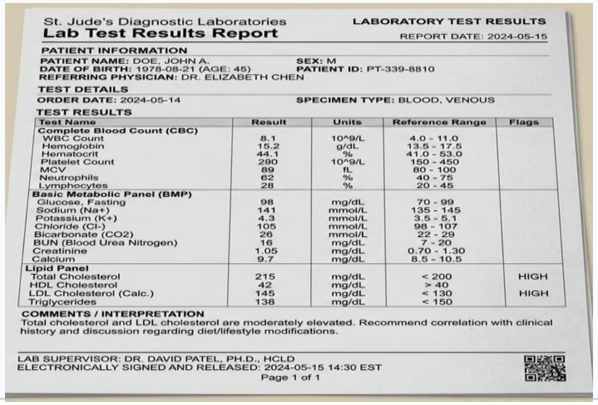
8.4
FKGL (target ≤ 8)
Readability is the known shortfall to improve

BIOMISTRAL TRAINING VS VALIDATION LOSS

System Demonstration & Interface Showcase

INTELLIGENT MEDICAL REPORT ANALYSIS SYSTEM

1 Image



St. Jude's Diagnostic Laboratories
Lab Test Results Report

LABORATORY TEST RESULTS
REPORT DATE: 2024-05-15

PATIENT INFORMATION

PATIENT NAME: DOE, JOHN A. **SEX:** M
DATE OF BIRTH: 1978-08-21 (AGE: 45) **PATIENT ID:** PT-339-8810
REFERRING PHYSICIAN: DR. ELIZABETH CHEN

TEST DETAILS

ORDER DATE: 2024-05-14 **SPECIMEN TYPE:** BLOOD, VENOUS

TEST RESULTS

Test Name	Result	Units	Reference Range	Flags
Complete Blood Count (CBC)				
WBC Count	8.1	10^9/L	4.0 - 11.0	
Hemoglobin	15.2	g/dL	13.5 - 17.5	
Hematocrit	44.1	%	41.0 - 53.0	
Platelet Count	280	10^9/L	150 - 450	
MCV	89	fL	80 - 100	
Neutrophils	62	%	40 - 75	
Lymphocytes	28	%	20 - 45	
Basic Metabolic Panel (BMP)				
Glucose, Fasting	98	mg/dL	70 - 99	
Sodium (Na+)	141	mmol/L	135 - 145	
Potassium (K+)	4.3	mmol/L	3.5 - 5.1	
Chloride (Cl-)	105	mmol/L	98 - 107	
Bicarbonate (CO2)	26	mmol/L	22 - 29	
BUN (Blood Urea Nitrogen)	16	mg/dL	7 - 20	
Creatinine	1.05	mg/dL	0.70 - 1.30	
Calcium	9.7	mg/dL	8.5 - 10.5	
Lipid Panel				
Total Cholesterol	215	mg/dL	< 200	HIGH
HDL Cholesterol	42	mg/dL	> 40	
LDL Cholesterol (Calc.)	145	mg/dL	< 130	HIGH
Triglycerides	138	mg/dL	< 150	

COMMENTS / INTERPRETATION
Total cholesterol and LDL cholesterol are moderately elevated. Recommend correlation with clinical history and discussion regarding diet/lifestyle modifications.

LAB SUPERVISOR: DR. DAVID PATEL, PH.D., HCLD
ELECTRONICALLY SIGNED AND RELEASED: 2024-05-15 14:30 EST
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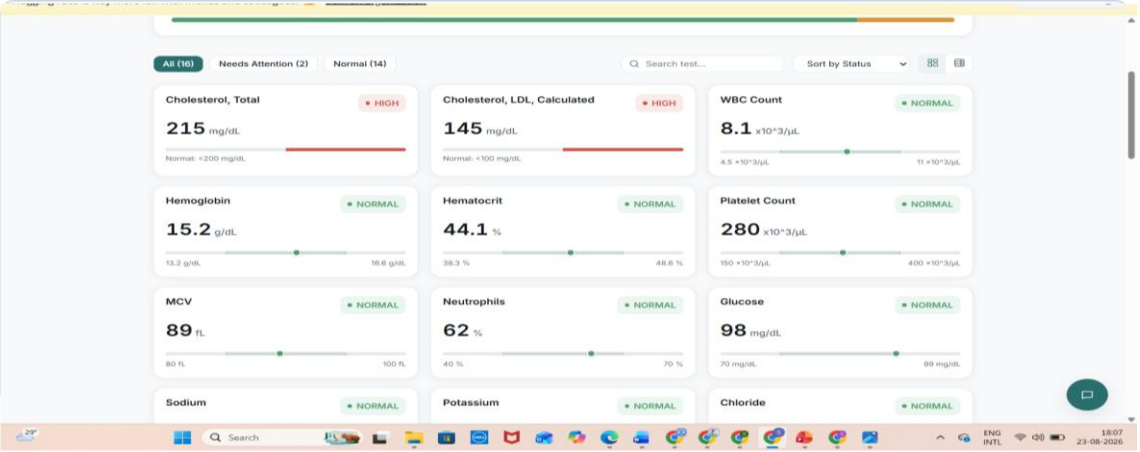
2 Dashboard



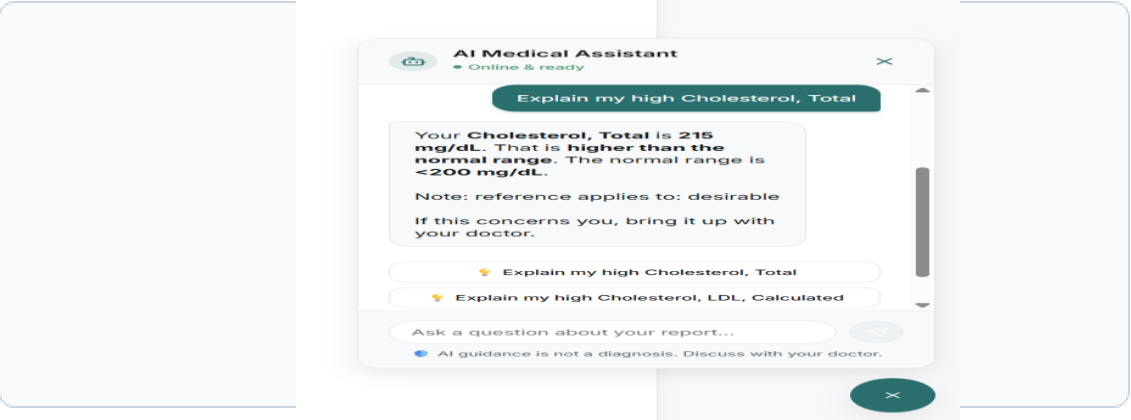
3 Explanations

Interactive Explanation

- **Hemoglobin (15.2 g/dL, Normal):** Hemoglobin carries oxygen in your blood. This level is normal.
- **Hematocrit (44.1 %, Normal):** Hematocrit measures the amount of blood that is made up of red blood cells. This result is normal.
- **Platelet Count (280.0 x10^3/µL, Normal):** Platelets help your blood clot to stop bleeding. This count is normal.
- **MCV (89.0 fL, Normal):** MCV measures the size of your red blood cells. This value is normal.
- **Neutrophils (62.0 %, Normal):** Neutrophils are white blood cells that fight infection. This percentage is normal.
- **Glucose (98.0 mg/dL, Normal):** Glucose is a type of sugar in your blood. This level is normal.
- **Sodium (141.0 mmol/L, Normal):** Sodium helps control the amount of water in your body. This level is normal.
- **Potassium (4.3 mmol/L, Normal):** Potassium is a mineral that helps your body work properly. This level is normal.
- **Chloride (105.0 mmol/L, Normal):** Chloride helps keep the balance of fluids and acids in your body. This level is normal.
- **Bicarbonate (26.0 mmol/L, Normal):** Bicarbonate helps keep the balance of acids and bases in your body. This level is normal.
- **Bun (16.0 mg/dL, Normal):** Bun is a type of nitrogen that helps your body work properly. This level is normal.
- **Creatinine (1.05 mg/dL, Normal):** Creatinine is a waste product that your kidneys remove from your blood. This level is normal.
- **Cholesterol, Total (215.0 mg/dL, High):** Cholesterol is a type of fat that can build up in your blood. A high level may suggest a problem with your heart or blood vessels, so it is worth checking

4 Q&A

Analytical Overview



Deployment & Privacy Architecture

ENTERPRISE SECURITY & COMPLIANCE

Deployment Infrastructure

- **GPU-enabled Deployment:** Inference backend deployed on **Hugging Face Spaces** for high-throughput BioMistral processing.
- **Local Execution Support:** Full end-to-end pipeline capable of running locally on suitable GPU hardware.
- **Self-Hosted Models:** ClinicalBERT & BioMistral run on dedicated endpoints without third-party LLM dependencies.

Privacy & Data Flow

- **Zero Persistent Storage:** Medical reports processed entirely in-memory and never stored permanently.
- **Transient Lifespan:** Data retained strictly for the duration of OCR, NER extraction, and explanation.
- **Isolated Transmission:** Data routes exclusively to the dedicated inference backend - never sent to public APIs.

Medical Safety & Guardrails

- **Non-Diagnostic Design:** System provides analytical explanations only; explicitly defers medical diagnosis to clinicians.
- **Medication Guardrails:** Clear warnings against altering or stopping prescribed medication without a physician.
- **Emergency Deferral:** High-risk or emergency indicators trigger immediate prompts for professional clinical care.

Architecture Comparison Matrix

Approach	Entity F1	Hallucination Risk	Data Privacy	Decision Rationale
Pretained ClinicalBERT)	**8.20%**	None	High	Not Selected
Heuristic Regex	**24.73%**	None	High	Not Selected
★ **Our Hybrid Pipeline**	99.91%	ZERO HALLUCINATION	TRANSIENT PROCESSING	SELECTED PIPELINE


Artificial Intelligence in
Healthcare
Presenter Name
The Future AI in Healthcare
Lorem Ipsum
AI Opportunity Analysis
AI Trends in Healthcare

THANK YOU


BLOOD TEST